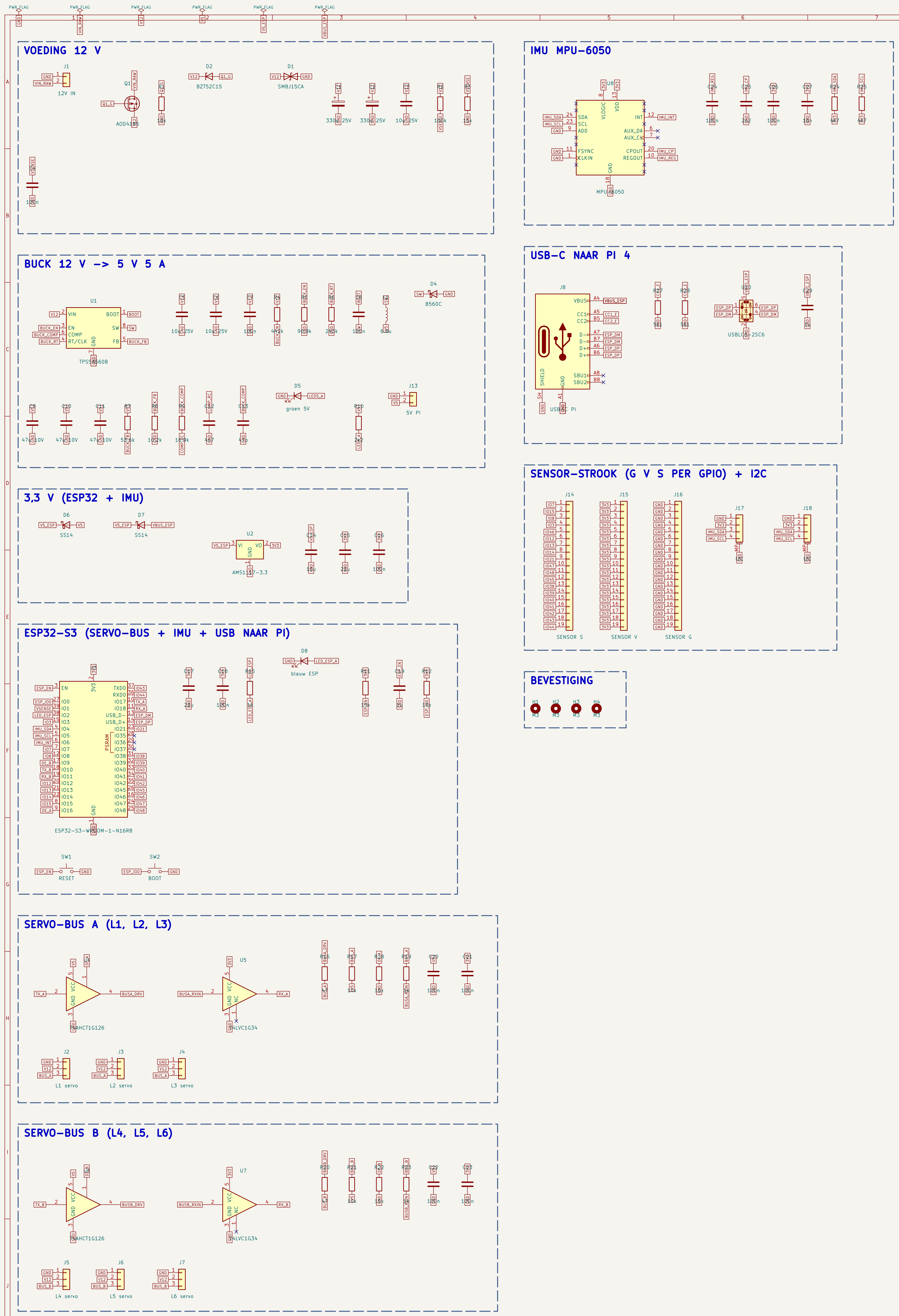
VOEDING 12 V
J1
12V IN
Q1
AOD4185
D2
BZT52C15
D1
SMBJ15CA
330u 25V
330u 25V
10u 25V
100n
IMU MPU-6050
U8
MPU-6050
BUCK 12 V -> 5 V 5 A
U1
TPS54560B
D4
B560C
10u 25V
10u 25V
100n
47u 10V
47u 10V
47u 10V
D5
green 5V
J13
5V PI
USB-C NAAR PI 4
J8
USB-C Pi
USBLC6-2SC6
SENSOR-STROOK (G V S PER GPIO) + I2C
J14
J15
J16
J17
J18
SENSOR S
SENSOR V
SENSOR G
3,3 V (ESP32 + IMU)
D6
SS14
D7
SS14
U2
AMS1117-3.3
BEVESTIGING
H1
H2
H3
H4
M3
M3
M3
M3
ESP32-S3 (SERVO-BUS + IMU + USB NAAR PI)
ESP32-S3-WROOM-1-N16R8
D8
blauw ESP
SW1
RESET
SW2
BOOT
SERVO-BUS A (L1, L2, L3)
U5
74AHCT1G126
74LVC1G34
J2
J3
J4
L1 servo
L2 servo
L3 servo
SERVO-BUS B (L4, L5, L6)
U7
74AHCT1G126
74LVC1G34
J5
J6
J7
L4 servo
L5 servo
L6 servo

ARACHNE-18 MINI BRAIN v1.0 – ESP32-S3 + MPU-6050 hexapod servo controller for a Raspberry Pi 4 (generated by tools/build_sch.py from tools/design.py)
Servo connectors: 1 = GND, 2 = +12 V, 3 = DATA (Feetech STS half-duplex bus, 1 Mbaud). CHECK the pin order against your cable before plugging in!
Bus A = legs L1 L2 L3 (ESP32 UART1: TX IO17, RX IO18, DE IO16). Bus B = legs L4 L5 L6 (UART2: TX IO10, RX IO11, DE IO9). DE = RTS in RS485 half-duplex mode.
Pi 4 link: USB-C (board) <-> USB-A (Pi) cable on the ESP32-S3 native USB -> /dev/ttyACM0 on the Pi. Pi VBUS only feeds the ESP32 (through D7), never the servos.
5V PI pads: buck output (5 V, 3 A for the Pi 4) -> Pi GPIO pin 2/4 (+5V) and 6 (GND), or a cut USB-C lead into the Pi power socket.
Buck = TI TPS54560B datasheet design (fig. 32): 400 kHz, 5.0 V 5 A. UVLO start 6.5 V.

ESP32-S3 + MPU-6050, 6 servo-bus legs, USB to Pi 4
Kobe Van Robaeys – maakleerplek
Sheet: /
File: arachne_mini.kicad_sch
Title: ARACHNE-18 mini brain board
Size: A1
Date: 2026-10-18
Rev: 1.0
KiCad E.D.A. 10.0.3
Id: 1/1